

MAT 320 HW06

Aditya Pahuja

Due 10/16

Problem 1. Prove that $x = \cos(x)$ for some $x \in (0, \pi/2)$.

Solution. Consider the function $f(x) = \cos(x) - x$. Assuming that $\cos(x)$ and x are both continuous on $[0, \pi/2]$, we can see that $f(x)$ is also continuous on $[0, \pi/2]$. Now, since

$$f(0) = \cos(0) - 0 = 1 > 0, \quad f(\pi/2) = \cos(\pi/2) - \pi/2 = -\pi/2 < 0,$$

the intermediate value theorem says that there exists an $x_0 \in [0, \pi/2]$ such that $f(x_0) = 0$. Clearly $f(0)$ and $f(\pi/2)$ are not zero, so $x_0 \neq 0$ and $x_0 \neq \pi/2$, thus $x_0 \in (0, \pi/2)$, which means that we have found a value of x (namely x_0) in the open interval $(0, \pi/2)$ such that

$$f(x) = \cos(x) - x = 0 \iff \cos(x) = x,$$

as desired. □

Problem 2. Show that a polynomial $P(x)$ of odd degree has at least one zero.

Solution. Let $P(x) = a_0 + a_1x + a_2x^2 + \cdots + a_nx^n$ where n is an odd integer and the a_i are real numbers such that $a_n \neq 0$.

Claim — There exist real numbers r_+ , r_- such that $P(r_+) > 0$ and $P(r_-) < 0$.

Proof. Consider the expression

$$\frac{P(x)}{x^n} = a_n + \frac{a_{n-1}}{x} + \cdots + \frac{a_1}{x^{n-1}} + \frac{a_0}{x^n} = a_n + \sum_{k=0}^{n-1} \frac{a_k}{x^{n-k}}.$$

Since

$$\lim_{x \rightarrow \infty} \frac{c}{x^k} = 0$$

for any real c and positive integer k , we can say that

$$\lim_{x \rightarrow \infty} \frac{P(x)}{x^n} = a_n + \lim_{x \rightarrow \infty} \sum_{k=0}^{n-1} \frac{a_k}{x^{n-k}} = a_n,$$

as the terms with x in the denominator grow arbitrarily small when x increases. Similarly, we have

$$\lim_{x \rightarrow \infty} \frac{P(-x)}{x^n} = \lim_{x \rightarrow \infty} -a_n + \sum_{k=0}^{n-1} \frac{a_k \cdot (-1)^k}{x^{n-k}} = -a_n,$$

where we use the fact that n is odd to conclude that $a_n(-x)^n = -a_nx^n$.

Now, suppose without loss of generality that $a_n > 0$ and let $\varepsilon = a_n$. By the definition of limit, the first limit tells us that there exists an $N_1 \in \mathbb{N}$ such that $x > N_1$ implies

$\left| \frac{P(x)}{x^n} - a_n \right| < \varepsilon$. Similarly, the second limit posits the existence of an $N_2 \in \mathbb{N}$ such that $x > N_2$ implies that $\left| \frac{P(-x)}{x^n} + a_n \right| < \varepsilon$. Equivalently, the inequalities rewrite as

$$0 = a_n - \varepsilon < \frac{P(x)}{x^n} < a_n + \varepsilon < 2a_n, \quad -2a_n = -a_n - \varepsilon < \frac{P(-x)}{x^n} < -a_n + \varepsilon = 0.$$

Since x^n is positive whenever $x > \max(N_1, N_2)$, we can say that $P(x)$ is positive and $P(-x)$ is negative for any choice of $x > \max(N_1, N_2)$. Then, we can take $r_+ = x$ and $r_- = -x$ to get the claim. $\square$

To finish, since $P(r_+)$ is positive and $P(r_-)$ is negative, we can use the fact that polynomials are continuous to conclude that, by the intermediate value theorem, there is an r_0 between r_- and r_+ for which $P(r_0) = 0$. $\square$

Problem 3. Prove that a continuous function $f : \mathbb{R} \rightarrow \mathbb{R}$ that takes on no value more than twice takes some value exactly once.

Solution. Let a and b be real numbers such that $a < b$ and $f(a) = f(b) = L$; if no such a and b exist, then f only takes each value at most once, so we are immediately done by picking any value in the image of f . By the Weierstrass theorem, there exists u and v such that, for all $x \in [a, b]$,

$$f(u) \leq f(x) \leq f(v).$$

Now, $f(u)$ and $f(v)$ cannot both be equal to L , as otherwise we would get $f(x) = L$ for all $x \in [a, b]$, and there are more than two values of x in $[a, b]$, which contradicts our assumption about f . So, suppose WLOG that $f(u) < L$ (as opposed to $f(v) > L$).

I claim that f takes on the value $m = f(u)$ exactly once. Suppose for the sake of contradiction that $f(x) = m$ has two solutions, say u and another real number w . Assume WLOG $u < w$ and consider two cases:

- $w \in [a, b]$. In this case, we know that $f(x) \geq m$ for all $x \in [a, b]$, so in particular $f(x) \geq m$ for all $x \in [u, w]$. The Weierstrass theorem tells us that, more specifically, f attains some maximum M on $[u, w]$; that is, there exists a real $t \in [u, w]$ such that $m \leq f(x) \leq f(t) = M$ when $x \in [u, w]$.

Now, let r be a real number in the interval $(m, \min(L, M))$. I will find three values x_1 , x_2 , and x_3 such that $f(x_1) = f(x_2) = f(x_3) = r$. First, observe that

$$f(u) = f(w) < r < f(a) = f(b),$$

so, by the intermediate value theorem on f over the intervals $[a, u]$ and $[w, b]$, there exists $x_1 \in [a, u]$ and $x_2 \in [w, b]$ such that $f(x_1) = f(x_2) = r$. Note that $r \neq m$ means that x_1 and x_2 are distinct from u and w , so really we can say $x_1 \in [a, u)$ and $x_2 \in (w, b]$. For the third value, we see that, by the intermediate value theorem, there exists x_3 within the interval $[u, t]$ such that $f(x_3) = r \in [m, M] = [f(u), f(t)]$.

This means r is achieved by f more than twice, which is a contradiction.

- $w \notin [a, b]$. Our assumption that $w > u$ means that $w > b$. Pick an r in the open interval (m, L) . Then, by intermediate value theorem, the following values exist:
 - $x_1 \in [a, u]$ satisfying $f(x_1) = r$, since $r \in [f(u), f(a)] = [m, L]$.
 - $x_2 \in [u, b]$ satisfying $f(x_2) = r$, since $r \in [f(u), f(b)] = [m, L]$.

◦ $x_3 \in [b, w]$ satisfying $f(x_3) = r$, since $r \in [f(w), f(b)] = [m, L]$.

Moreover, no two of the x_i can be equal, as $x_1 = x_2 = u$ would imply $r = m$ and $x_2 = x_3$ would imply $r = L$, both of which contradict $r \in (m, L)$ (and obviously $x_1 < x_2 < x_3$ so $x_1 \neq x_3$). Thus, we again have found an r that is achieved by f more than twice, which is a contradiction.

To conclude, $f(x) = m$ must only have one real solution u , as desired. $\square$

Problem 4. Let d and d' be two equivalent metrics on X . Show that:

- a set $A \subseteq X$ is open in (X, d) if and only if it is open in (X, d') .
- a set $A \subseteq X$ is closed in (X, d) if and only if it is closed in (X, d') .
- a function $f: X \rightarrow \mathbb{R}$ is continuous with respect to the metric d if and only if it is continuous with respect to the metric d' .
- a sequence (x_n) of points in X converges to some $\ell \in X$ with respect to the metric d if and only if it converges with respect to ℓ with respect to d' .

Solution. We can rephrase the equivalence of d and d' as saying that there exist positive reals a and b such that

$$a \leq \frac{d(x, y)}{d'(x, y)} \leq b$$

for all $x, y \in X$. That is, the ratio of the distances between x and y in the two metrics is bounded within an interval with positive real endpoints. This lets us see that equivalence of metrics is symmetric — if $\frac{d}{d'}$ is in $[a, b]$, then $\frac{d'}{d}$ is always in $[\frac{1}{b}, \frac{1}{a}]$. Thus, it suffices to prove just the forward direction (assuming d has the property) for each of the statements, as the argument in the backward direction will be identical.

Openness. If $A \subseteq X$ is open with respect to d , then, for each $x \in A$, there exists an ε such that $B(x, \varepsilon_x) \subseteq A$. That is, for each $y \in X$,

$$d(x, y) < \varepsilon_x \implies y \in A.$$

Using the equivalence of d and d' ,

$$a \cdot d'(x, y) \leq d(x, y) < \varepsilon_x \implies y \in A.$$

This can be rephrased as

$$d'(x, y) < \frac{\varepsilon_x}{a} \implies y \in A,$$

meaning that, for each $x \in A$, we have found an open ball $B(x, \frac{\varepsilon_x}{a})$ (defined in the metric space (X, d')) centered at x that is a subset of A . Thus, A is also open with respect to d' (and as mentioned above, the converse holds because we can just swap d and d' in the proof).

Closedness. If $A \subseteq X$ is closed with respect to d , then we can equivalently say that its complement $X \setminus A$ is open with respect to d . By the previous section, this is equivalent to $X \setminus A$ being open with respect to d' , which is equivalent to its complement A being closed with respect to d' .

Continuity. Suppose $f: X \rightarrow \mathbb{R}$ is continuous in (X, d) . Then, for each $c \in X$ and $\varepsilon > 0$, there exists δ such that

$$d(x, c) < \delta \implies d_{\mathbb{R}}(f(x), f(c)) = |f(x) - f(c)| < \varepsilon$$

(where I am assuming that $\mathbb{R}$ is equipped with the standard absolute value metric $d_{\mathbb{R}}(x, y) = |x - y|$). The condition $d(x, c) < \delta$ implies that

$$a \cdot d'(x, c) \leq d(x, c) < \delta \implies d'(x, c) < \frac{\delta}{a}.$$

That is, given ε and c , if δ is the quantity mentioned above, then we can say that there exists $\delta' = \frac{\delta}{a}$ such that

$$d'(x, c) < \delta' \implies d_{\mathbb{R}}(f(x), f(c)) = |f(x) - f(c)| < \varepsilon.$$

Convergence. Supposing that a sequence (x_n) of points in X converges to some ℓ with respect to d , we wish to show that (x_n) converges to ℓ with respect to d' as well, meaning that for any $\varepsilon > 0$, we want to show the existence of an N such that

$$n > N \implies d'(x_n, \ell) < \varepsilon.$$

By definition of convergence (with respect to d), there exists an N_0 such that

$$n > N_0 \implies d(x_n, \ell) < a\varepsilon,$$

since $a\varepsilon$ is a positive real number. In particular $n > N_0$ implies that

$$a \cdot d'(x_n, \ell) \leq d(x_n, \ell) < a\varepsilon \implies d'(x_n, \ell) < \varepsilon,$$

so taking $N = N_0$ suffices to show that (x_n) converges with respect to d' , too. $\square$

Problem 5. Let $X = \mathbb{R}^n$. For $\mathbf{x} = (x_1, \dots, x_n)$ and $\mathbf{y} = (y_1, \dots, y_n)$ in X set

$$d_1(\mathbf{x}, \mathbf{y}) = \sum_{i=1}^n |x_i - y_i| \quad \text{and} \quad d_2(\mathbf{x}, \mathbf{y}) = \max_{i=1, \dots, n} |x_i - y_i|.$$

Show that d_1 and d_2 are both distance functions on X . Show that the standard metric on X we defined during class

$$d(\mathbf{x}, \mathbf{y}) = \sqrt{\sum_{i=1}^n |x_i - y_i|^2}$$

is equivalent to both d_1 and d_2 .

Solution. First, we show that d_1 and d_2 are metrics on X . Indeed, both are clearly functions $X \times X \rightarrow \mathbb{R}_{\geq 0}$ where $\mathbb{R}_{\geq 0}$ is the set of nonnegative reals (because absolute values and sums of absolute values are always nonnegative). We verify the other properties of metrics individually. First, for d_1 :

- $d_1(\mathbf{x}, \mathbf{y}) = 0$ iff $\mathbf{x} = \mathbf{y}$. First, clearly if $\mathbf{x} = \mathbf{y}$ then $|x_i - y_i| = 0$ for all i , hence their sum $d_1(\mathbf{x}, \mathbf{y})$ is also zero.

Now, we prove the converse statement that if $\mathbf{x} \neq \mathbf{y}$, then the distance between them is not zero. To show this, we observe that $|x_i - y_i| \geq 0$ for all i , so, if, for some k , $|x_k - y_k|$ is positive (which $x_k \neq y_k$), then we are guaranteed to have

$$\sum_{i=1}^n |x_i - y_i| \geq |x_k - y_k| > 0,$$

meaning $d_1(\mathbf{x}, \mathbf{y}) \neq 0$.

- $d_1(\mathbf{x}, \mathbf{y}) = d_1(\mathbf{y}, \mathbf{x})$. This is fairly simple, as, because $|x_i - y_i| = |y_i - x_i|$, we have

$$d_1(\mathbf{x}, \mathbf{y}) = \sum_{i=1}^n |x_i - y_i| = \sum_{i=1}^n |y_i - x_i| = d_1(\mathbf{y}, \mathbf{x}).$$

- Triangle inequality. Let $\mathbf{z} = (z_1, \dots, z_n)$. Using the standard triangle inequality $|a - b| + |b - c| \geq |a - c|$,

$$d_1(\mathbf{x}, \mathbf{z}) + d_1(\mathbf{z}, \mathbf{y}) = \sum_{i=1}^n |x_i - z_i| + |z_i - y_i| \geq \sum_{i=1}^n |x_i - y_i| = d_1(\mathbf{x}, \mathbf{y}).$$

For d_2 :

- $d_2(\mathbf{x}, \mathbf{y}) = 0$ iff $\mathbf{x} = \mathbf{y}$. First, clearly if $\mathbf{x} = \mathbf{y}$ then $|x_i - y_i| = 0$ for all i , hence their max $d_2(\mathbf{x}, \mathbf{y})$ is also zero.

Now, we prove the converse statement that if $\mathbf{x} \neq \mathbf{y}$, then the distance between them is not zero. To show this, we observe that $|x_i - y_i| \geq 0$ for all i , so, if, for some k , $|x_k - y_k|$ is positive (which $x_k \neq y_k$), then we are guaranteed to have

$$\max_{i=1, \dots, n} |x_i - y_i| \geq |x_k - y_k| > 0,$$

meaning $d_2(\mathbf{x}, \mathbf{y}) \neq 0$.

- $d_1(\mathbf{x}, \mathbf{y}) = d_1(\mathbf{y}, \mathbf{x})$. This is fairly simple, as, because $|x_i - y_i| = |y_i - x_i|$, we have

$$d_2(\mathbf{x}, \mathbf{y}) = \max_{i=1, \dots, n} |x_i - y_i| = \max_{i=1, \dots, n} |y_i - x_i| = d_2(\mathbf{y}, \mathbf{x}).$$

- Triangle inequality. Let $\mathbf{z} = (z_1, \dots, z_n)$. Using the standard triangle inequality $|a - b| + |b - c| \geq |a - c|$,

$$\begin{aligned} d_2(\mathbf{x}, \mathbf{y}) &= \max_{i=1, \dots, n} |x_i - y_i| \\ &\leq \max_{i=1, \dots, n} (|x_i - z_i| + |z_i - y_i|) \\ &\leq \max_{i=1, \dots, n} \left(|x_i - z_i| + \max_{i=1, \dots, n} |z_i - y_i| \right) \\ &= \max_{i=1, \dots, n} |x_i - z_i| + \max_{i=1, \dots, n} |z_i - y_i| \\ &= d_2(\mathbf{x}, \mathbf{z}) + d_2(\mathbf{z}, \mathbf{y}). \end{aligned}$$

To be clear, the second line is at most as large as the third line because the maximum of $|z_i - y_i|$ across all i is always at least as large as any one of the $|z_i - y_i|$, and the third line is equal to the fourth line because we are just separating the constant $\max |z_i - y_i|$ from the outer max.

It remains to show that d_1 and d_2 are equivalent to d . We prove the following inequality, which clearly shows that the metrics are all equivalent.

$$\frac{d_1(\mathbf{x}, \mathbf{y})}{n} \leq d_2(\mathbf{x}, \mathbf{y}) \leq d(\mathbf{x}, \mathbf{y}) \leq \sqrt{n} \cdot d_2(\mathbf{x}, \mathbf{y}) \leq \sqrt{n} \cdot d_1(\mathbf{x}, \mathbf{y}).$$

This splits into four inequalities in the obvious way; we prove them individually. First,

$$\frac{d_1(\mathbf{x}, \mathbf{y})}{n} = \frac{\sum_{i=1}^n |x_i - y_i|}{n} \leq \frac{\sum_{i=1}^n \max_{i=1, \dots, n} |x_i - y_i|}{n} = \frac{n \cdot d_2(\mathbf{x}, \mathbf{y})}{n} = d_2(\mathbf{x}, \mathbf{y}).$$

Second,

$$d_2(\mathbf{x}, \mathbf{y}) = \sqrt{\max_{i=1, \dots, n} |x_i - y_i|^2} \leq \sqrt{\sum_{i=1}^n |x_i - y_i|^2} = d(\mathbf{x}, \mathbf{y}).$$

Third,

$$d(\mathbf{x}, \mathbf{y}) = \sqrt{\sum_{i=1}^n |x_i - y_i|^2} \leq \sqrt{n \cdot \max_{i=1, \dots, n} |x_i - y_i|^2} = \sqrt{n} \cdot d_2(\mathbf{x}, \mathbf{y}).$$

Fourth,

$$\sqrt{n} \cdot d_2(\mathbf{x}, \mathbf{y}) = \sqrt{n} \cdot \max_{i=1, \dots, n} |x_i - y_i| \leq \sqrt{n} \cdot \sum_{i=1}^n |x_i - y_i| = \sqrt{n} \cdot d_1(\mathbf{x}, \mathbf{y}).$$

In particular we have that

$$\frac{d_1(\mathbf{x}, \mathbf{y})}{n} \leq d(\mathbf{x}, \mathbf{y}) \leq \sqrt{n} \cdot d_1(\mathbf{x}, \mathbf{y}),$$

which shows that d and d_1 are equivalent, and

$$d_2(\mathbf{x}, \mathbf{y}) \leq d(\mathbf{x}, \mathbf{y}) \leq \sqrt{n} \cdot d_2(\mathbf{x}, \mathbf{y}),$$

which shows that d and d_2 are equivalent, as desired. $\square$

Problem 6. Prove that if (X, d_X) and (Y, d_Y) are metric spaces then their Cartesian product $X \times Y$ is also a metric space when equipped with the metric

$$d_{X \times Y}((x, y), (x', y')) = d_X(x, x') + d_Y(y, y').$$

Solution. Since $d_X(x, x')$ and $d_Y(y, y')$ are always nonnegative, so is their sum; therefore $d_{X \times Y}$ is always nonnegative. Moreover,

$$d_X(x, x') + d_Y(y, y') = 0$$

if and only if $d_X(x, x')$ and $d_Y(y, y')$ are both zero, since if either is nonzero then the sum is positive; therefore, the sum $d_{X \times Y}((x, y), (x', y'))$ is zero if and only if $x = x'$ and $y = y'$. We also have

$$\begin{aligned} d_{X \times Y}((x, y), (x', y')) &= d_X(x, x') + d_Y(y, y') \\ &= d_X(x', x) + d_Y(y', y) \\ &= d_{X \times Y}((x', y'), (x, y)), \end{aligned}$$

and, for the triangle inequality, if we have a third point $(\hat{x}, \hat{y})$, we can use the triangle inequalities for d_X and d_Y to get

$$\begin{aligned} d_{X \times Y}((x, y), (\hat{x}, \hat{y})) + d_{X \times Y}((\hat{x}, \hat{y}), (x', y')) &= (d_X(x, \hat{x}) + d_Y(y, \hat{y})) + (d_X(\hat{x}, x') + d_Y(\hat{y}, y')) \\ &= (d_X(x, \hat{x}) + d_X(\hat{x}, x')) + (d_Y(y, \hat{y}) + d_Y(\hat{y}, y')) \\ &\geq d_X(x, x') + d_Y(y, y') \\ &= d_{X \times Y}((x, y), (x', y')). \end{aligned} \quad \square$$

Problem 7. Since $\mathbb{R}^4 = \mathbb{R}^3 \times \mathbb{R} = \mathbb{R}^2 \times \mathbb{R}^2 = \mathbb{R} \times \mathbb{R}^3$ there are three different product metrics on $\mathbb{R}^4$. Write down an explicit formula for each one of those. What is the product metric on $\mathbb{R} \times \mathbb{R} \times \cdots \times \mathbb{R} = \mathbb{R}^n$?

Solution. Let $\mathbf{x} = (x_1, x_2, x_3, x_4)$ and $\mathbf{y} = (y_1, y_2, y_3, y_4)$. The product metric on $\mathbb{R}^3 \times \mathbb{R}$ is

$$d_{\mathbb{R}^3 \times \mathbb{R}}(\mathbf{x}, \mathbf{y}) = \sqrt{|x_1 - y_1|^2 + |x_2 - y_2|^2 + |x_3 - y_3|^2 + |x_4 - y_4|^2}.$$

The product metric on $\mathbb{R}^2 \times \mathbb{R}^2$ is

$$d_{\mathbb{R}^2 \times \mathbb{R}^2}(\mathbf{x}, \mathbf{y}) = \sqrt{|x_1 - y_1|^2 + |x_2 - y_2|^2} + \sqrt{|x_3 - y_3|^2 + |x_4 - y_4|^2}.$$

The product metric on $\mathbb{R} \times \mathbb{R}^3$ is

$$d_{\mathbb{R} \times \mathbb{R}^3}(\mathbf{x}, \mathbf{y}) = |x_1 - y_1| + \sqrt{|x_2 - y_2|^2 + |x_3 - y_3|^2 + |x_4 - y_4|^2}.$$

For $\mathbb{R} \times \mathbb{R} \times \cdots \times \mathbb{R} = \mathbb{R}^n$, if $\mathbf{x} = (x_1, \dots, x_n)$ and $\mathbf{y} = (y_1, \dots, y_n)$, then

$$d_{\mathbb{R} \times \mathbb{R} \times \cdots \times \mathbb{R}} = \sum_{i=1}^n d_{\mathbb{R}}(x_i, y_i) = \sum_{i=1}^n |x_i - y_i|. \quad \square$$

Problem 8. Show that the product metric $d_{\mathbb{R}^n \times \mathbb{R}^m}$ on $\mathbb{R}^n \times \mathbb{R}^m = \mathbb{R}^{n+m}$ is equivalent to the standard metric

$$d(\mathbf{x}, \mathbf{y}) = \sqrt{\sum_{i=1}^{n+m} |x_i - y_i|^2}.$$

Solution. We write d' instead of $d_{\mathbb{R}^n \times \mathbb{R}^m}$. We will prove the inequality

$$\frac{1}{\sqrt{2}} d'(\mathbf{x}, \mathbf{y}) \leq d(\mathbf{x}, \mathbf{y}) \leq d'(\mathbf{x}, \mathbf{y}),$$

which by definition implies the metrics are equivalent. Define

$$p = \sum_{i=1}^n |x_i - y_i|^2, \quad q = \sum_{i=n+1}^{n+m} |x_i - y_i|^2$$

so that

$$d'(\mathbf{x}, \mathbf{y}) = \sqrt{p} + \sqrt{q}, \quad d(\mathbf{x}, \mathbf{y}) = \sqrt{p + q}.$$

Then, squaring gives the equivalent inequality

$$\frac{p + q + 2\sqrt{pq}}{2} \leq p + q \leq p + q + 2\sqrt{pq}.$$

The left inequality reduces to

$$\sqrt{pq} \leq \frac{p + q}{2} \iff 4pq \leq p^2 + 2pq + q^2 \iff 0 \leq p^2 - 2pq + q^2 = (p - q)^2$$

which holds because all squares are nonnegative (or this is just the AM-GM inequality for two variables). The right inequality is trivial because $2\sqrt{pq} \geq 0$. Thus, the metrics are equivalent. $\square$

Problem 9. In class we defined $\ell^2 \subseteq \mathbb{R}^\infty$ to be the set consisting of all infinite sequences $\mathbf{x} = (x_1, x_2, x_3, \dots)$ such that

$$\sum_{n=1}^{\infty} x_n^2 < +\infty.$$

During class, for two sequences $\mathbf{x}$ and $\mathbf{y}$ in ℓ^2 we defined their distance to be

$$d(\mathbf{x}, \mathbf{y}) = \sqrt{\sum_{n=1}^{\infty} |x_n - y_n|^2}.$$

Explain why this series converges.

Solution. We want to show that

$$\sum_{n=1}^{\infty} |x_n - y_n|^2 = \sum_{n=1}^{\infty} x_n^2 + y_n^2 - 2x_n y_n$$

converges, given that (x_n) and (y_n) are sequences in ℓ^2 .

Claim — The series

$$\sum_{n=1}^{\infty} x_n y_n$$

converges absolutely.

Proof. Since $\mathbf{x}, \mathbf{y} \in \ell^2$, we know that

$$\sum_{n=1}^{\infty} x_n^2, \quad \sum_{n=1}^{\infty} y_n^2$$

both converge. Moreover, we know that

$$x_n^2 - 2|x_n y_n| + y_n^2 = (|x_n| - |y_n|)^2 \geq 0 \iff x_n^2 + y_n^2 \geq 2|x_n y_n| \geq |x_n y_n|$$

for all natural numbers n , so we can see by limit comparison that

$$\sum_{n=1}^{\infty} |x_n y_n| \leq \sum_{n=1}^{\infty} x_n^2 + y_n^2 < +\infty,$$

as desired. □

The claim in particular tells us that

$$\sum_{n=1}^{\infty} x_n y_n$$

converges (absolute convergence implies convergence), so we can say that

$$\sum_{n=1}^{\infty} x_n^2 + y_n^2 - 2x_n y_n = \sum_{n=1}^{\infty} x_n^2 + \sum_{n=1}^{\infty} y_n^2 - 2 \sum_{n=1}^{\infty} x_n y_n < +\infty,$$

hence the metric on ℓ^2 is well-defined. □

Problem 10. Define $\ell^1 \subseteq \mathbb{R}^\infty$ to be the set of all infinite sequences $\mathbf{x} = (x_1, x_2, x_3, \dots)$ that give rise to an absolutely convergent series. Show that ℓ^1 is a metric space with distance

$$d(\mathbf{x}, \mathbf{y}) = \sum_{n=1}^{\infty} |x_n - y_n|.$$

Show that $\ell^1 \subseteq \ell^2$. Find a sequence that is in ℓ^2 but not in ℓ^1 (i.e. show that the inclusion is strict).

Solution. Clearly $d \geq 0$ over all of its inputs, and the distance function is symmetric because

$$d(\mathbf{x}, \mathbf{y}) = \sum_{n=1}^{\infty} |x_n - y_n| = \sum_{n=1}^{\infty} |y_n - x_n| = d(\mathbf{y}, \mathbf{x}).$$

We also see that $d(\mathbf{x}, \mathbf{y}) = 0$ if and only if

$$\sum_{n=1}^{\infty} |x_n - y_n| = 0.$$

Since each $|x_n - y_n|$ term is nonnegative, if any of the terms is positive then the sum will be positive, thus the only way the distance between two sequences is zero is if $|x_n - y_n| = 0$ for all n , i.e. $\mathbf{x} = \mathbf{y}$.

To finish, the triangle inequality follows directly from the one-dimensional version:

$$\begin{aligned} d(\mathbf{x}, \mathbf{z}) + d(\mathbf{z}, \mathbf{y}) &= \sum_{n=1}^{\infty} |x_n - z_n| + \sum_{n=1}^{\infty} |z_n - y_n| \\ &= \sum_{n=1}^{\infty} |x_n - z_n| + |z_n - y_n| \\ &\geq \sum_{n=1}^{\infty} |x_n - y_n| \\ &= d(\mathbf{x}, \mathbf{y}). \end{aligned}$$

Thus, we have shown that $d: \ell^1 \times \ell^1 \rightarrow [0, +\infty)$ is zero if and only if its arguments are positive, is symmetric, and satisfies the triangle inequality, meaning it is a metric on ℓ^1 .

Now, we show $\ell^1 \subseteq \ell^2$. If $\mathbf{x} \in \ell^1$, then $|x_n| < 1$ eventually, and then $x_n^2 < |x_n| < 1$ eventually, say when $n > N$. By limit comparison,

$$\sum_{n>N} x_n^2 < \sum_{n>N} |x_n| < +\infty,$$

and $x_1^2, \dots, x_N^2$ don't affect the convergence of the series, so we can conclude that $\sum x_n^2$ converges. Additionally, the sequence given by $x_n = \frac{1}{n}$ is in ℓ^2 but not ℓ^1 , as $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges while $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges. $\square$